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Semiconductor device having channel regions distributed on two opposite sides of a gate electrode

Abstract

Embodiments relate to a semiconductor device and a forming method. The semiconductor device includes: a substrate; a memory array positioned on the substrate and at least including memory cells spaced along a first direction, each of the memory cells including a transistor, the transistor including a gate electrode, channel regions distributed on two opposite sides of the gate electrode along a third direction, and a source region and a drain region distributed on two opposite sides of each of the channel regions along a second direction, the first direction and the third direction being directions parallel to a top surface of the substrate, the first direction intersecting with the third direction, and the second direction being a direction perpendicular to the top surface of the substrate; and a word line extending along the first direction and continuously electrically connected to the gate electrodes spaced along the first direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Chinese Patent Application No. 202210743273.X, titled “SEMICONDUCTOR DEVICE AND METHOD FOR FORMING SAME” and filed to the State Patent Intellectual Property Office on Jun. 28, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the field of semiconductor fabrication technology, and more particularly, to a semiconductor device and a method for forming the same.

BACKGROUND

(3) As a semiconductor apparatus commonly used in electronic devices such as computers, Dynamic Random Access Memory (DRAM) comprises a plurality of memory cells, where each of the plurality of memory cells generally includes a transistor and a capacitor. A gate of the transistor is electrically connected to a word line, a source of the transistor is electrically connected to a bit line, and a drain of the transistor is electrically connected to the capacitor. A word line voltage of the word line can control on or off of the transistor, such that data information stored in the capacitor can be read or written into the capacitor by means of the bit line.

(4) With continuous miniature of sizes of semiconductor devices such as the DRAM, the semiconductor devices have developed from two-dimensional structures to three-dimensional

stacked structures. However, due to limitations of processes for fabricating the semiconductor devices and fabrication yield of the semiconductor devices, it is difficult to further reduce the sizes of the semiconductor devices.

(5) Therefore, how to simplify the processes for fabricating the semiconductor devices, increase the yield of the semiconductor devices, and improve an integration level of the semiconductor devices is a technical problem to be solved urgently at present.

SUMMARY

(6) A semiconductor device and a method for forming the same are provided by some embodiments of the present disclosure.

(7) According to some embodiments, the present disclosure provides a semiconductor device, which includes: a substrate; a memory array positioned on the substrate and at least comprising a plurality of memory cells spaced along a first direction, each of the plurality of memory cells comprising a transistor, the transistor comprising a gate electrode, channel regions distributed on two opposite sides of the gate electrode along a third direction, and a source region and a drain region distributed on two opposite sides of each of the channel regions along a second direction, the first direction and the third direction being directions parallel to a top surface of the substrate, the first direction intersecting with the third direction, and the second direction being a direction perpendicular to the top surface of the substrate; and a word line extending along the first direction and being continuously electrically connected to the plurality of gate electrodes spaced along the first direction.

(8) According to some other embodiments, the present disclosure further provides a method for forming the semiconductor device, including: providing an initial substrate; and forming, in the initial substrate, a substrate, a memory array positioned above the substrate, and a word line, where the memory array at least comprises a plurality of memory cells spaced along a first direction, each of the plurality of memory cells comprising a transistor, the transistor comprising a gate electrode, channel regions distributed on two opposite sides of the gate electrode along a third direction, and a source region and a drain region distributed on two opposite sides of each of the channel regions along a second direction; the word line extends along the first direction and is continuously electrically connected to the plurality of gate electrodes spaced along the first direction, where the first direction and the third direction are directions parallel to a top surface of the substrate, the first direction intersecting with the third direction, and the second direction being a direction perpendicular to the top surface of the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram showing a first structure of a semiconductor device according to an embodiment of the present disclosure;

(2) FIGS. 2A to 2B are two schematic structural diagrams of a capacitor in the semiconductor device in FIG. 1;

(3) FIG. 3 is a schematic diagram showing a second structure of the semiconductor device according to an embodiment of the present disclosure;

(4) FIGS. 4A to 4B are two schematic structural diagrams of the capacitor in the semiconductor device in FIG. 3;

(5) FIG. 5 is a schematic diagram showing a third structure of the semiconductor device according to an embodiment of the present disclosure;

(6) FIG. 6A to FIG. 6B are two schematic structural diagrams of the semiconductor device in FIG. 5;

(7) FIG. 7 is a schematic top-down view of the capacitor according to an embodiment of the

present disclosure;

(8) FIGS. **8A** to **8B** are two schematic structural diagrams showing connection between a gate electrode and a word line according to an embodiment of the present disclosure;

(9) FIG. **9** is a flowchart of a method for forming the semiconductor device according to an embodiment of the present disclosure; and

(10) FIGS. **10** to **19** are schematic structural diagrams showing main processes in a process of forming the semiconductor device according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(11) Embodiments of a semiconductor device and a method for forming the same provided by the present disclosure will be described in detail below with reference to the accompanying drawings.

(12) This embodiment provides a semiconductor device. FIG. **1** is a schematic diagram showing a first structure of the semiconductor device according to an embodiment of the present disclosure.

FIGS. **2A** to **2B** are two schematic structural diagrams of a capacitor in the semiconductor device in FIG. **1**. FIG. **3** is a schematic diagram showing a second structure of the semiconductor device according to an embodiment of the present disclosure. FIGS. **4A** to **4B** are two schematic structural diagrams of the capacitor in the semiconductor device in FIG. **3**. FIG. **5** is a schematic diagram showing a third structure of the semiconductor device according to an embodiment of the present disclosure. FIG. **6A** to FIG. **6B** are two schematic structural diagrams of the semiconductor device in FIG. **5**. FIG. **7** is a schematic top-down view of the capacitor according to an embodiment of the present disclosure. FIGS. **8A** to **8B** are two schematic structural diagrams showing connection between a gate electrode and a word line according to an embodiment of the present disclosure. As shown in FIG. **1**, FIGS. **2A** to **2B**, FIG. **3**, FIGS. **4A** to **4B**, FIG. **5**, FIGS. **6A** to **6B**, FIG. **7**, and FIGS. **8A** to **8B**, the semiconductor device includes: a substrate; a memory array positioned on the substrate and at least comprising a plurality of memory cells spaced along a first direction **D1**, each of the plurality of memory cells comprising a transistor, the transistor comprising a gate electrode **11**, channel regions **14** distributed on two opposite sides of the gate electrode **11** along a third direction **D3**, and a source region **13** and a drain region **15** distributed on two opposite sides of each of the channel regions **14** along a second direction **D2**, the first direction **D1** and the third direction **D3** being directions parallel to a top surface of the substrate, the first direction **D1** intersecting with the third direction **D3**, and the second direction **D2** being a direction perpendicular to the top surface of the substrate; and a word line **80** extending along the first direction **D1** and being continuously electrically connected to the plurality of gate electrodes **11** spaced along the first direction **D1**.

(13) The semiconductor device in this embodiment may be, but is not limited to, a dynamic random access memory (DRAM). For example, the substrate may be, but is not limited to, a silicon substrate. This embodiment is described by taking an example where the substrate is the silicon substrate. In other embodiments, the substrate may be a semiconductor substrate such as gallium nitride, gallium arsenide, gallium carbide, silicon carbide, or silicon on insulator (SOI). The substrate is configured to support device structures thereon. The top surface of the substrate refers to a surface of the substrate facing towards the memory array. The memory array may include a plurality of the memory cells arranged in an array along the first direction **D1** and the third direction **D3**, where the third direction **D3** is a direction parallel to the top surface of the substrate, and the first direction **D1** intersects the third direction **D3**. In one embodiment, the memory array may further include the plurality of memory cells arranged in a three-dimensional array along the first direction **D1**, the second direction **D2** and the third direction **D3**, thereby further increasing integration degree and storage capacity of the semiconductor device.

(14) The memory cell includes a transistor, which includes an active pillar **10** and the gate electrode **11**. In an embodiment, the active pillar **10** may extend along the second direction **D2**. The active pillar **10** includes the source region **13**, the channel region **14**, and the drain region **15** stacked in sequence along the second direction **D2**. The gate electrode **11** is embedded inside the active pillar

10, such that the channel regions **14** are distributed on two opposite sides of the gate electrode **11** along the third direction **D3** to form a double-channel structure, such that fabrication difficulty of the semiconductor device is reduced while improving electrical performance of the semiconductor device. The word line **80** extends along the first direction and is continuously electrically connected to the plurality of gate electrodes **11** spaced along the first direction **D1**. That is, an extension direction of the word line **80** is orthogonal to an extension direction of the active pillar **10** to further improve the integration level of the semiconductor device. In one embodiment, the memory cell further includes a gate dielectric layer **19** between the gate electrode **11** and the channel region **14**. A material of the gate dielectric layer **19** may be, but not limited to, an oxide material (such as silicon dioxide).

(15) In some embodiments, a width of the gate electrode **11** along the third direction **D3** is greater than or equal to that of the word line **80** along the third direction **D3**.

(16) In an example, as shown in FIG. **8A**, to reduce adverse effects of a short channel effect on the performance of the transistor, the width of the gate electrode **11** along the third direction **D3** may be greater than that of the word line **80** along the third direction **D3**. In another example, as shown in FIG. **8B**, to further simplify manufacturing processes of the semiconductor device, the width of the gate electrode **11** along the third direction **D3** may be equal to that of the word line **80** along the third direction **D3**. The intersection mentioned in this embodiment may be a vertical intersection (i.e., orthogonal intersection) or an oblique intersection.

(17) In some embodiments, the semiconductor device further includes: a bit line **12** positioned below the transistor and extending along the third direction **D3**, where the bit line **12** is electrically connected to the source region **13**.

(18) In some embodiments, the bit line **12** extends along the third direction **D3** and is continuously electrically connected to the source regions **13** in the plurality of memory cells spaced along the third direction **D3**. In an embodiment, the bit line **12** may be positioned inside the substrate to form a buried bit line structure, thereby further reducing the size of the semiconductor device. In another embodiment, the bit line **12** may also be positioned above the substrate to simplify a process for forming the bit line **12** and reduce an internal resistance of the bit line **12**. A material of the bit line **12** may be metal material or a combination of metal silicide and the metal material, which may be selected by those skilled in the art according to actual needs.

(19) In some embodiments, one of the memory cells further includes: a capacitor structure positioned above the transistor along the second direction **D2**, where the capacitor structure is electrically connected to the drain region **15**; a first isolation layer positioned between the source region **13** and the gate electrode **11**; and a second isolation layer **18** positioned between the capacitor structure and the gate electrode **11**.

(20) In some embodiments, the first isolation layer may be a single-layer or multi-layer structure, and is configured to electrically isolate the source region **13** from the gate electrode **11**. In one embodiment, the first isolation layer includes a first sub isolation layer **21** and a second sub isolation layer **22** positioned above the first sub isolation layer **21** to further reduce a parasitic capacitance effect of the memory cell. A material of the first sub isolation layer **21** may be, but is not limited to, an oxide material (such as silicon dioxide), and a material of the second sub isolation layer **22** may be, but is not limited to, a nitride material (such as silicon nitride). The second isolation layer **18** may also be a single-layer or multi-layer structure (for example, including oxide layers and nitride layers alternately stacked along the second direction **D2**), to electrically isolate the gate electrode **11** from the capacitor structure.

(21) In some embodiments, as shown in FIG. **1**, the capacitor structure includes: a capacitor contact **17** positioned above the gate electrode **11**, where the drain region **15** is distributed around a side wall of the capacitor contact **17**; and a capacitor **16** positioned above the capacitor contact **17** and electrically connected to the capacitor contact **17**.

(22) In some embodiments, as shown in FIG. **2A**, the capacitor **16** includes: a lower electrode layer

22 being columnar and extending along the second direction D2, where the lower electrode layer 22 is electrically connected to the drain region 15; a dielectric layer 23 covering a surface of the lower electrode layer 22; and an upper electrode layer 24 covering a surface of the dielectric layer 23.

(23) FIG. 1(a) and FIG. 1(b) are schematic cross-sectional views of the same semiconductor device along different directions. In some embodiments, as shown in FIGS. 1 and 2A, the capacitor structure includes the capacitor contact 17 positioned above the second isolation layer 18, and the drain region 15 is distributed around a side wall of the capacitor contact 17 and is electrically connected to the capacitor contact 17. The capacitor 16 is positioned above the capacitor contact 17 and is in electrical connection with the capacitor contact 17. By arranging the capacitor contact 17, a contact resistance between the capacitor 16 and the drain region 15 can be reduced, and connection stability between the capacitor 16 and the drain region 15 can be enhanced. In an embodiment, a material of the capacitor contact 17 may be a metal material (such as tungsten) or a conductive material such as titanium nitride (TiN).

(24) In an example, the capacitor 16 includes the lower electrode layer 22 covering a top surface of the capacitor contact 17 and having a columnar cross section, the dielectric layer 23 covering the surface of the lower electrode layer 22 and the top surface of the active pillar 10, and the upper electrode layer 24 covering the surface of the dielectric layer 23. In an embodiment, the semiconductor device further includes a common electrode layer covering a surface of the upper electrode layer 24, and the plurality of capacitor structures in the semiconductor device share the common electrode layer. That is, the common electrode layer continuously covers the upper electrode layers 24 of the plurality of capacitors 16. A material of the lower electrode layer 22 and a material of the upper electrode layer 24 may be same or may be different. In an embodiment, to save fabrication costs of the semiconductor device, the material of the lower electrode layer 22 and the material of the upper electrode layer 24 are the same, for example, one or a combination of two or more of TiN, titanium oxide (TiO), and the metal material. The dielectric layer 23 is formed from a material having a higher dielectric constant. In an embodiment, a material of the dielectric layer 23 may be oxide, oxynitride, silicon oxide, or silicon oxynitride containing any one or two or more of hafnium, ruthenium, zirconium, and tantalum. A material of the common electrode layer may be, but is not limited to, polysilicon.

(25) In an example, as shown in FIG. 7, for the capacitor 16 including the lower electrode layer 22 having the columnar cross section, a radius R1 of the lower electrode layer 22 may be 4 nm to 50 nm, a radius R2 of the capacitor 16 may be 10 nm to 70 nm, and a length of the capacitor 16 along the second direction D2 may be 200 nm to 1,500 nm.

(26) In some other embodiments, the capacitor as shown in FIG. 2B includes: a lower electrode layer 22 extending along the second direction D2 and being electrically connected to the drain region 15, where a cross section of the lower electrode layer 22 is U-shaped; a dielectric layer 23 covering a surface of the lower electrode layer 22; and an upper electrode layer 24 covering a surface of the dielectric layer 23.

(27) In some embodiments, the lower electrode layer 22 includes a bottom surface and side surfaces extending along the second direction D2 and distributed around an outer periphery of a bottom surface of the lower electrode layer 22, such that the cross section of the lower electrode layer 22 is U-shaped to increase a surface area of the lower electrode layer 22, thereby increasing a surface area of the capacitor, and finally increasing capacitance of the capacitor 16. The bottom surface of the U-shaped lower electrode layer 22 covers the top surface of the capacitor contact 17 and is electrically connected to the capacitor contact 17, the dielectric layer 23 covers the surface of the lower electrode layer 22, and the upper electrode layer 24 covers the surface of the dielectric layer 23. In an embodiment, the semiconductor device further includes a common electrode layer covering the surface of the upper electrode layer 24, and the plurality of capacitor structures in the semiconductor device share one common electrode layer. That is, the common electrode layer

continuously covers the upper electrode layers **24** of the plurality of capacitors **16**.

(28) In some other embodiments, as shown in FIG. **3**, the capacitor structure includes: a capacitor **16** positioned above the gate electrode **11**, where the drain region **15** is distributed around a side wall of part of the capacitor **16**.

(29) FIG. **3(a)** and FIG. **1(b)** are schematic cross-sectional views of the same semiconductor device along different directions. In an example, as shown in FIG. **3** and FIG. **4A**, the capacitor **16** includes the lower electrode layer **22** having the columnar cross section, the dielectric layer **23** covering the surface of the lower electrode layer **22**, and the upper electrode layer **24** covering the surface of the dielectric layer **23**. The drain region **15** is distributed around a side wall of part of the lower electrode layer **22** and is electrically connected to the lower electrode layer **22** to increase a contact area between the drain region **15** and the lower electrode layer **22**, and to reduce a contact resistance between the drain region **15** and the lower electrode layer **22**. The dielectric layer **23** covers a surface of the lower electrode layer **22** extending out of the active pillar **10** and a top surface of the drain region **15**.

(30) In another example, as shown in FIGS. **3** and **4B**, the capacitor **16** includes the lower electrode layer **22** with the U-shaped cross section, the dielectric layer **23** covering the surface of the lower electrode layer **22**, and the upper electrode layer **24** covering the surface of the dielectric layer **23**. The lower electrode layer **22** includes the bottom surface and the side surfaces extending along the second direction **D2** and distributed around the outer periphery of the bottom surface of the lower electrode layer **22**, such that the cross section of the lower electrode layer **22** is U-shaped to increase the surface area of the lower electrode layer **22**, thereby increasing the surface area of the capacitor, and finally increasing the capacitance of the capacitor **16**. The drain region **15** is distributed around the side wall of part of the lower electrode layer **22** and is electrically connected to the lower electrode layer **22** to increase the contact area between the drain region **15** and the lower electrode layer **22**, and to reduce the contact resistance between the drain region **15** and the lower electrode layer **22**.

(31) In some embodiments, as shown in FIG. **5**, the capacitor structure includes: a capacitor terminal **50** covering the top surface of the drain region **15**; and the capacitor **16** positioned above the capacitor terminal **50** and being electrically connected to the capacitor terminal **50**.

(32) FIG. **5(a)** and FIG. **1(b)** are schematic cross-sectional views of the same semiconductor device along different directions. In an example, as shown in FIG. **5**, the capacitor structure includes the capacitor terminal **50** positioned on the top surface of the drain region **15**, and the drain region **15** is in electrical connection with the capacitor terminal **50**. The capacitor **16** is positioned on a top surface of the capacitor terminal **50** and is in electrical connection with the capacitor terminal **50**. By arranging the capacitor terminal **50**, a contact resistance between the capacitor **16** and the drain region **15** can be reduced. In an embodiment, a material of the capacitor terminal **50** may be polysilicon. The cross section of the lower electrode layer **22** in the capacitor **16** may be columnar as shown in FIG. **6A** to simplify a process for fabricating the capacitor **16**. In some embodiments, the cross section of the lower electrode layer **22** in the capacitor **16** may be U-shaped as shown in FIG. **6B** to increase capacitance of the capacitor **16**.

(33) This embodiment also provides a method for forming the semiconductor device. FIG. **9** is a flowchart of the method for forming the semiconductor device according to an embodiment of the present disclosure. FIGS. **10** to **19** are schematic structural diagrams showing main processes in a process of forming the semiconductor device according to some embodiments of the present disclosure, where FIG. **10** is a schematic top-down view of the semiconductor device formed in this embodiment, and FIGS. **11** to **19** are schematic cross-sectional views showing main processes in a process of forming the semiconductor device at an a-a' position and a b-b' position in FIG. **10**, to clearly show the formation process of the semiconductor device. The structure of the semiconductor device formed in this embodiment may be seen in FIG. **1**, FIGS. **2A** to **2B**, FIG. **3**, FIGS. **4A** to **4B**, FIG. **5**, FIGS. **6A** to **6B**, FIG. **7**, and FIGS. **8A** to **8B**. As shown in FIG. **1**, FIGS.

2A to 2B, FIG. 3, FIGS. 4A to 4B, FIG. 5, FIGS. 6A to 6B, FIG. 7, FIGS. 8A to 8B, and FIGS. 9 to 19, the method for forming the semiconductor device includes following steps.

(34) Step S91, providing an initial substrate.

(35) In some embodiments, the initial substrate may be, but is not limited to, a silicon substrate. This embodiment is described by taking an example where the initial substrate is the silicon substrate. In other embodiments, the initial substrate may be a semiconductor substrate such as gallium nitride, gallium arsenide, gallium carbide, silicon carbide, or silicon on insulator (SOI).

(36) Step S92, forming, in the initial substrate, a substrate **100**, a memory array positioned above the substrate **100**, and a word line **80**, where the memory array at least comprises a plurality of memory cells spaced along the first direction D1. Each of the plurality of memory cells comprises a transistor, where the transistor comprises a gate electrode **11**, channel regions **14** distributed on two opposite sides of the gate electrode **11** along the third direction D3, and a source region **13** and a drain region **15** distributed on two opposite sides of each of the channel regions **14** along the second direction D2. The word line **80** extends along the first direction D1 and is continuously electrically connected to the plurality of gate electrodes **11** spaced along the first direction D1, where the first direction D1 and the third direction D3 are directions parallel to the top surface of the substrate **110**, the first direction D1 intersects with the third direction D3, and the second direction D2 is a direction perpendicular to the top surface of the substrate **110**, as shown in FIG. 1, FIG. 10, and FIG. 16.

(37) In some embodiments, the step of forming, in the initial substrate, the substrate **100**, the memory array positioned above the substrate **100**, and the word line **80** includes: etching the initial substrate to form a plurality of first trenches and a plurality of active pillars **10** spaced at least along the first direction D1, where a portion of the initial substrate remained below the first trench serves as the substrate **100**, as shown in FIG. 11; defining, in the active pillars **10**, a channel region **14**, and a source region **13** and a drain region **15** distributed on two opposite sides of the channel region **14** along the second direction D2, as shown in FIG. 1; forming a second trench **120** penetrating through the drain region **15** and the channel region **14** along the second direction D2, as shown in FIG. 12; and forming, in the second trench **120**, the gate electrode **11** and the word line **80** extending along the first direction D1 and being continuously electrically connected to the plurality of gate electrodes **11** spaced along the first direction D1.

(38) In some embodiments, the initial substrate is etched to form the plurality of first trenches for dividing the initial substrate into the plurality of active pillars **10** arrayed along the first direction D1 and the third direction D3, where the third direction D3 is a direction parallel to a top surface of the substrate **110**, and the third direction D1 intersects with the first direction D1. The intersection in this embodiment may be an oblique intersection or a vertical intersection. To facilitate subsequent formation of the gate electrode **11** in the active pillar **10**, a width of the active pillar **10** along the third direction D3 may be 30 nm to 40 nm, a width of the active pillar **10** along the first direction D1 may be 15 nm to 20 nm, and a width of the first trench along the first direction D1 and a width of the first trench along the third direction D3 may be 15 nm to 20 nm. Next, after a protective layer covering the active pillar **10** is formed on a side wall of the first trench, the initial substrate is continued to be etched along the first trench to form grooves positioned below the first trench and continuously communicating with the plurality of first trenches spaced along the third direction D3. Next, a bit line material is filled in the groove to form a bit line **12** extending along the third direction D3, as shown in FIG. 10 and FIG. 11. Next, an insulating dielectric material such as oxide (e.g., silicon dioxide) is filled in the first trench to form a third isolation layer **111**, which is configured to electrically isolate adjacent active pillars **10**.

(39) Next, the active pillar **10** may be etched by means of a double pattern photolithographic process to form a second trench **120** penetrating through the drain region **15** and the channel region **14** along the second direction D2, as shown in FIG. 12.

(40) In some embodiments, the step of forming, in the second trench **120**, the gate electrode **11** and

the word line **80** extending along the first direction **D1** and being continuously electrically connected to the plurality of gate electrodes **11** spaced along the first direction **D1** includes: forming a first isolation layer on a bottom surface of the second trench **120**; forming, on a side wall of the second trench **120**, a gate dielectric layer **19** covering the channel region **14**, as shown in FIG. **15**; and forming, in the second trench **120**, the gate electrode **11** covering the gate dielectric layer **19** and the first isolation layer, and simultaneously forming the word line **80** extending along the first direction **D1** and being continuously electrically connected to the plurality of gate electrodes **11** spaced along the first direction **D1**.

(41) In some embodiments, after forming, in the second trench **120**, the gate electrode **11** and the word line **80** extending along the first direction **D1** and being continuously electrically connected to the plurality of gate electrodes **11** spaced along the first direction **D1**, the method further includes following steps of: forming, in the second trench **120**, a second isolation layer **18** covering a top surface of the gate electrode **11**, as shown in FIG. **16**; and forming, above the second isolation layer **18**, the capacitor structure electrically connected to the drain region **15**.

(42) In some embodiments, the step of forming, above the second isolation layer **18**, the capacitor structure electrically connected to the drain region **15** includes: forming, in the second trench **120**, a capacitor contact **17** positioned on a surface of the second isolation layer **18** and electrically connected to the drain region **15**, as shown in FIG. **17**; and forming, above the capacitor contact **17**, a capacitor **16** electrically connected to the capacitor contact **17**, as shown in FIG. **18**.

(43) For example, an initial first sub isolation layer **131** covering an inner wall of the second trench **120**, and an initial second sub isolation layer **130** filling up the second trench **120** and covering a surface of the initial first sub isolation layer **131** may be formed by means of an atomic layer deposition process, as shown in FIG. **13**. A material of the initial first sub isolation layer **131** may be, but not limited to, oxide (such as silicon dioxide), and a material of the initial second sub isolation layer **130** may be, but not limited to, nitride (such as silicon nitride). Next, part of the initial first sub isolation layer **131** and part of the initial second sub isolation layer **130** are etched back, a portion of the initial first sub isolation layer **131** remained at a bottom of the second trench **120** serves as a first sub isolation layer **21**, a portion of the initial second sub isolation layer **130** remained above the first sub isolation layer **21** serves as a second sub isolation layer **20**, and the first sub isolation layer **21** and the second sub isolation layer **20** together serve as the first isolation layer, as shown in FIG. **14**. Next, a gate dielectric layer **19** covering a side wall of the second trench **120** is formed in the second trench **120** by means of an in situ steam generation (ISSG) process or a rapid thermal oxidation (RTO) process, as shown in FIG. **15**.

(44) Next, a conductive material such as TiN or tungsten is deposited in the plurality of second trenches **120** to form initial word lines filling up the second trenches **120**, extending along the first direction **D1** and continuously covering the plurality of gate dielectric layers **19** spaced along the first direction **D1**. The initial word line and the gate dielectric layer **19** are etched back to expose the drain region **15** in the active pillar, only the gate dielectric layer **19** on the surface of the channel region **14** is retained, a portion of the remaining initial word line surrounded by the channel region **14** serves as the gate electrode **11**, and the initial word line positioned between adjacent gate electrodes **11** serves as the word line **80**. Next, an insulating dielectric material such as nitride (e.g., silicon nitride) is deposited in the second trench **120** to form a second isolation layer **18** covering the top surface of the gate electrode **11**, and a top surface of the second isolation layer **18** is positioned below the top surface of the second trench **120**, as shown in FIG. **16**.

(45) Next, a conductive material such as TiN is deposited on the inner wall of the second trench **120** to form a diffusion barrier layer **171** positioned above the second isolation layer **18**. A conductive material such as tungsten is deposited in the second trench **120** to form the capacitor contact **17** covering a surface of the diffusion barrier layer **171** and filling up the second trench **120**. Next, a plurality of capacitors **16** electrically connected to the capacitor contacts **17** are formed above each of the capacitor contacts **17**, and a common electrode layer **25** continuously covering

the upper electrode layers of the plurality of capacitors **16** is formed, as shown in FIG. **18**.

(46) In some other embodiments, the step of forming, above the second isolation layer **18**, the capacitor structure electrically connected to the drain region **15** includes: forming a capacitor **16** on a surface of the second isolation layer **18**, where at least part of the capacitor **16** is positioned in the second trench **120**, as shown in FIG. **19**.

(47) In some embodiments, to simplify the fabrication processes and reduce the contact resistance between the capacitor **16** and the drain region **15**, after the second isolation layer **18** is formed, the capacitor contact **17** may not be formed, instead the capacitor **16** begins to be formed directly in the second trench **120**, such that the drain region **15** is distributed around the side wall of at least part of the capacitor **16**.

(48) According to the semiconductor device and the method for forming the same provided by some embodiments of the present disclosure, the channel regions are arranged on two opposite sides of the gate electrode to form a fully depleted gate electrode, and an arrangement direction of the source region, the channel regions, and a drain region in the transistor intersects with an extension direction of the word line. The semiconductor device is compact in structure, such that the integration level of the semiconductor device can be improved, the size of the semiconductor device can be reduced, and the yield of the semiconductor device can be increased. In addition, the gate electrode in some embodiments of the present disclosure is formed by directly filling a second trench after formation of the second trench in an active pillar, and the fabrication process is simple, thereby reducing fabrication difficulty of the semiconductor device.

(49) The above merely are embodiments of the present disclosure. It is to be pointed out that to those of ordinary skill in the art, various improvements and embellishments may be made without departing from the principles of the present disclosure, and these improvements and embellishments are also deemed to be within the scope of protection of the present disclosure.

Claims

1. A semiconductor device, comprising: a substrate; a memory array positioned on the substrate and at least comprising a plurality of memory cells spaced along a first direction, each of the plurality of memory cells comprising a transistor, the transistor comprising a gate electrode, channel regions distributed on two opposite sides of the gate electrode along a third direction, and a source region and a drain region distributed on two opposite sides of each of the channel regions along a second direction, the first direction and the third direction being directions parallel to a top surface of the substrate, the first direction intersecting with the third direction, and the second direction being a direction perpendicular to the top surface of the substrate; and a word line extending along the first direction and being continuously electrically connected to the plurality of gate electrodes spaced along the first direction.

2. The semiconductor device according to claim 1, wherein a width of the gate electrode along the third direction is greater than or equal to a width of the word line along the third direction.

3. The semiconductor device according to claim 1, further comprising: a bit line positioned below the transistor and extending along the third direction, the bit line being electrically connected to the source region.

4. The semiconductor device according to claim 1, wherein each of the plurality of memory cells further comprises: a capacitor structure positioned above the transistor along the second direction, the capacitor structure being electrically connected to the drain region; a first isolation layer positioned between the source region and the gate electrode; and a second isolation layer positioned between the capacitor structure and the gate electrode.

5. The semiconductor device according to claim 4, wherein the capacitor structure comprises: a capacitor contact positioned above the gate electrode, the drain region being distributed around a side wall of the capacitor contact; and a capacitor positioned above the capacitor contact and being

electrically connected to the capacitor contact.

6. The semiconductor device according to claim 5, wherein the capacitor comprises: a lower electrode layer being columnar and extending along the second direction, the lower electrode layer being electrically connected to the drain region; a dielectric layer covering a surface of the lower electrode layer; and an upper electrode layer covering a surface of the dielectric layer.

7. The semiconductor device according to claim 5, wherein the capacitor comprises: a lower electrode layer extending along the second direction and being electrically connected to the drain region, a cross section of the lower electrode layer being U-shaped; a dielectric layer covering a surface of the lower electrode layer; and an upper electrode layer covering a surface of the dielectric layer.

8. The semiconductor device according to claim 4, wherein the capacitor structure comprises: a capacitor positioned above the gate electrode, the drain region being distributed around a side wall of part of the capacitor.

9. The semiconductor device according to claim 4, wherein the capacitor structure comprises: a capacitor terminal covering a top surface of the drain region; and a capacitor positioned above the capacitor terminal and being electrically connected to the capacitor terminal.
